RAM BODEPUDI

+1(972)510-8164 | Dallas, TX | rammaruthyb@gmail.com | Ram Maruthy LinkedIn | **https://github.com/ram5112**

OBJECTIVE

Experienced AI and Machine Learning Engineer with comprehensive expertise in Generative AI, Machine Learning, and Data Science, delivering transformative solutions across healthcare, logistics, and enterprise applications. Proficient in leveraging cutting-edge technologies such as LangChain, Retrieval-Augmented Generation (RAG) frameworks, and advanced NLP models, including GPT-4, BERT, T5, and to develop robust and scalable AI-driven systems. Adept at integrating AI with structured and unstructured data, combining XGBoost for analytical insights with generative models for contextual narratives, enabling impactful, data-informed decision-making.Successfully implemented a RAG framework to streamline diagnostic report generation, reducing generation time by 45% while achieving 94% accuracy, validated by medical professionals during pilot testing. Enhanced healthcare workflows by fine-tuning domain-specific NLP models and building pipelines that comply with HIPAA and GDPR, ensuring security and scalability in cloud environments with Docker, Kubernetes, and AWS. In logistics, developed ML models predicting truck delays by analyzing historical route data, traffic patterns, and real-time weather conditions, improving delivery time accuracy and increasing logistics efficiency by over 30%.Proficient in end-to-end machine learning pipelines, from data ingestion and preprocessing to feature engineering using Hopsworks and AWS SageMaker, coupled with automation via CI/CD pipelines using GitHub Actions and AWS CodePipeline. Skilled in deployment of interactive dashboards with Flask, Streamlit, and React, facilitating real-time analytics and actionable insights for stakeholders.Adept at collaborating with cross-functional teams to translate business challenges into AI-powered solutions, backed by a strong foundation in programming (Python, C++, Java, SQL), cloud computing (AWS ECS, EKS), and frameworks (Django, TensorFlow, PyTorch). With a Master's in Computer Science and a proven track record of innovation, I am passionate about leveraging AI to tackle complex challenges and drive meaningful impact.

EDUCATION

Masters of Computer Science, University of Texas at Arlington **2022 - 2024**
Bachelor of Computer Science, Gandhi Institute of Technology and Management **2017 - 2021**

SKILLS

Programming Languages: Python, C++,Java,SQL, Javascript, Html, Css
Machine Learning: TensorFlow, PyTorch, Keras, scikit-learn,MLLib,MLFlow
Gen Ai and Deep learning Models: LLM-Langchain , Llama, Transformers(BERT, Huggingface)
Frameworks: Flask, Django, React,Streamlit
Data Analysis: Pandas, NumPy, Matplotlib, Seaborn
Integration and Reporting: SSIS, PowerBI, Tableau
Database: SQL Server, MySQL, PostgreSQL, Sqllite
Feature Store: Hopsworks, AWS SageMaker
Big Data and Cloud: AWS, ECS, EKS, Terraform
Version Control: Git, Github
Deployment and Scaling : Docker, Kubernetes

EXPERIENCE

Humana / CloudInfra
TX, USA

Gen AI Engineer **Jan 2024 - Present**

- Utilized Hugging Face Transformers to build and optimize a generative chatbot, leveraging pre-trained models and fine-tuning them to enhance conversational abilities for various use cases, including customer support and sales.
- Integrated neural networks and Support Vector Machines (SVM) within the chatbot's framework, improving the prediction of user intents and increasing the accuracy of responses, resulting in better user engagement and satisfaction.
- Designed and executed scalable data ingestion pipelines using AWS S3 and EC2, ensuring the chatbot can efficiently process and store large volumes of user interactions in real-time, while maintaining high performance under heavy loads.
- Applied advanced Natural Language Processing (NLP) techniques for real-time sentiment analysis and categorization of user feedback, enabling the chatbot to adapt its responses based on user emotions and preferences.
- Developed and fine-tuned NLP models with Hugging Face and Google Dialog Flow to enable the chatbot to understand context, handle complex queries, and provide highly accurate, relevant responses to user inputs.
- Engineered decision trees and ensemble methods to predict user intents with higher precision, minimizing response errors, and significantly improving the overall user experience by reducing miscommunication.
- Collaborated with cross-functional teams to integrate the chatbot with existing customer support systems, which streamlined operations, increased response efficiency, and reduced resolution times by 15

- Managed large-scale unstructured conversational data using NoSQL and PL/SQL databases, enabling seamless retrieval and storage for continuous learning and refinement of chatbot models.
- Automated data preprocessing and model retraining workflows by leveraging AWS EC2 instances, ensuring that the chatbot models remain up-to-date with fresh data and capable of adapting to evolving user interactions.
- Developed personalized recommendation algorithms that adapt chatbot interactions based on user history, preferences, and behavior patterns, driving higher user satisfaction and engagement on digital platforms.

Environment: Python, Hugging Face Transformers, PyTorch, TensorFlow, Jupyter Notebook,Vscode, Scikit-learn, Keras, Python, NumPy, Matplotlib, AWS SDK (Boto3), AWS Lambda, S3, EC2, Pandas,SpaCy, NLTK, BERT, Hugging Face Pipelines, AWS Comprehend,TensorFlow, Dialog Flow API, Hugging Face Transformer,Scikit-learn, XGBoost, Random Forest,REST APIs, Flask/Django, Postman

Cargill / Cognizant Technology Solution(Implementation partner)
Hyderabad, India

Machine Learning Engineer **Jan 2021 - June 2022**

- Developed machine learning models using Scikit-learn and Pandas to predict truck delays based on historical route data, weather conditions, and traffic patterns, improving delivery time accuracy and logistics efficiency.
- Built data pipelines for ingestion, cleaning, and transformation using Pandas and Hopsworks to accommodate continuous training on new data, ensuring the model remains up-to-date and adaptable to changing logistics patterns.
- Handled model performance improvements by fine-tuning hyperparameters, conducting error analysis, and incorporating additional features such as real-time traffic updates and weather forecasts, leading to improved prediction accuracy.
- Migrated the backend database from PostgreSQL to Amazon RDS, ensuring scalability and optimized data management, with seamless integration into existing systems.
- Utilized Hopsworks for feature engineering and data versioning, improving the efficiency and reproducibility of feature extraction for truck delay prediction models.
- Deployed truck delay prediction models on Amazon ECS with Docker containerization, enhancing scalability and operational efficiency for production environments.
- Built CI/CD pipelines using GitHub Actions and AWS CodePipeline for automating testing, deployment, and model updates, reducing manual intervention and ensuring consistent deployments.
- Integrated MLflow for model tracking and repository management, enabling version control, experiment tracking, and reproducible workflows.
- Implemented an interactive dashboard using Streamlit, deployed on Amazon EC2, to monitor and visualize truck delays and predicted arrival times, enhancing decision-making for logistics managers.
- Performed data cleaning and preprocessing using NumPy and Pandas to ensure accurate and high-quality data inputs from multiple sources, including GPS, weather APIs, and traffic data
- Developed RESTful APIs with Flask to provide third-party logistics platforms access to truck delay predictions, improving collaboration and delivery transparency. Additionally, integrated third-party APIs for real-time traffic and weather data, enhancing the model's ability to predict future delays accurately.

Environment: Python, SQL, Scikit-learn, Pandas, NumPy, MLflow, Matplotlib, Seaborn, Hopsworks, AWS Lambda, PostgreSQL, Amazon RDS, Amazon ECS, Docker, AWS EC2, AWS CodePipeline, GitHub Actions, Flask, RESTful APIs, third-party traffic and weather APIs, Streamlit, Jupyter Notebook, Vscode

Fractal Analytics / Internship
Hyderabad, India

Data Scientist **Apr 2019 - Dec 2020**

- Analyzed clinical trial data using Python and Pandas, identifying patterns to enhance research outcomes.
- Built machine learning models with Scikit-learn and TensorFlow for personalized medicine predictions.
- Developed real-time data pipelines with PostgreSQL and utilized Amazon EMR, EC2, and S3 for scalable data
- management.
- Designed predictive maintenance models and conducted NLP on medical literature to support drug development.
- Created data visualizations for stakeholders using Matplotlib and managed PostgreSQL databases for efficient storage.
- Integrated ML models into healthcare applications via Flask APIs, logging metrics with MLflow for monitoring.
- Leveraged GitHub for version control and optimized supply chain management using predictive algorithms.

Environment: Python, Pandas, Scikit-learn, TensorFlow, PostgreSQL, Amazon EMR, AWS EC2, AWS S3, Matplotlib, Flask, MLflow, GitHub, Jupyter Notebook, Vscode

PROJECTS

- Image Classifier of Motorbikes, Schooners, and Ships Using Convolutional Neural Networks
- Reviews Sentiment Analysis Using Support Vector Machines

CERTIFICATIONS

AWS Certified Developer Associate
Python Certified Programmer [PCEP] and IBM Applied AI Professional Certification